COMPUTER ENGINEERING

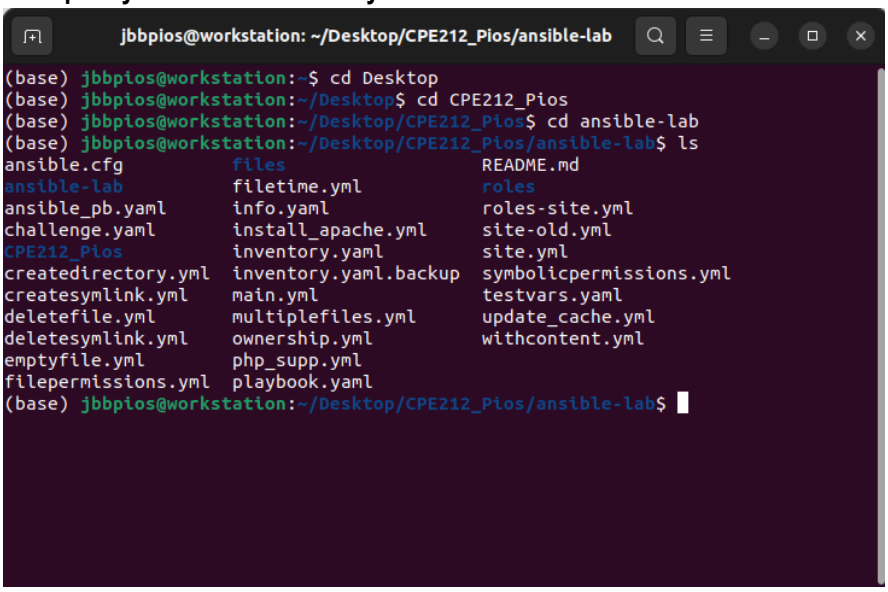
QUEZON CITY

COURSE CODE/TITLE:	CPE 212	**SECTION:**	CPE31S2
DATE PERFORMED:	September 17, 2026	**DATE SUBMITTED:**	September 17, 2026
NAME:	Pios, Joshua Paul B.	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

ACTIVITY NO. 8.1 **Install, Configure, and Manage Enterprise Availability Monitoring via Ansible**

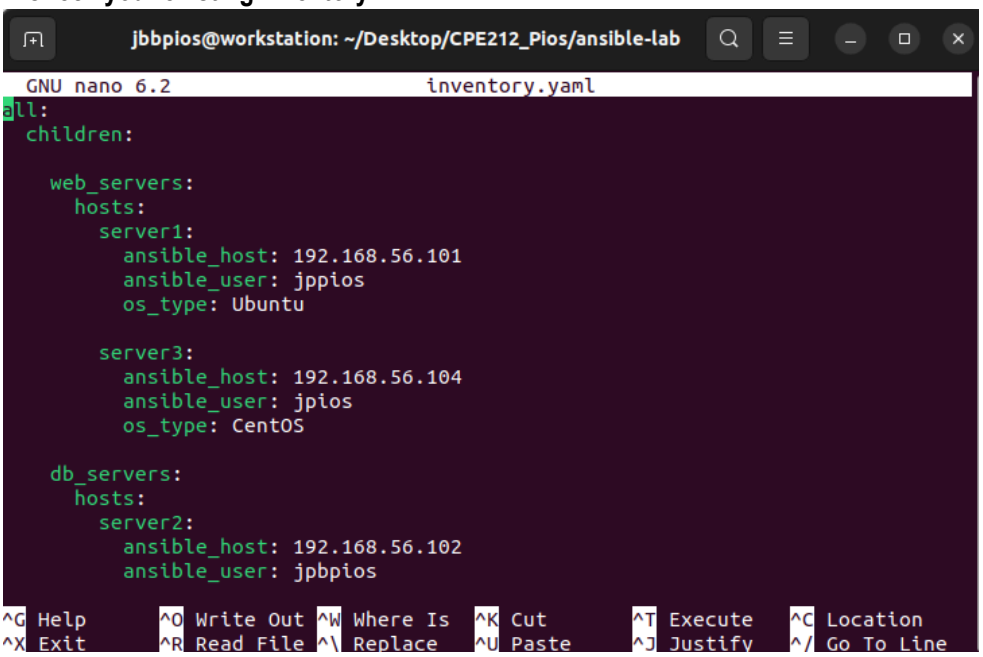
1. PROCEDURE OUTPUT

1. Prepare your Ansible Directory



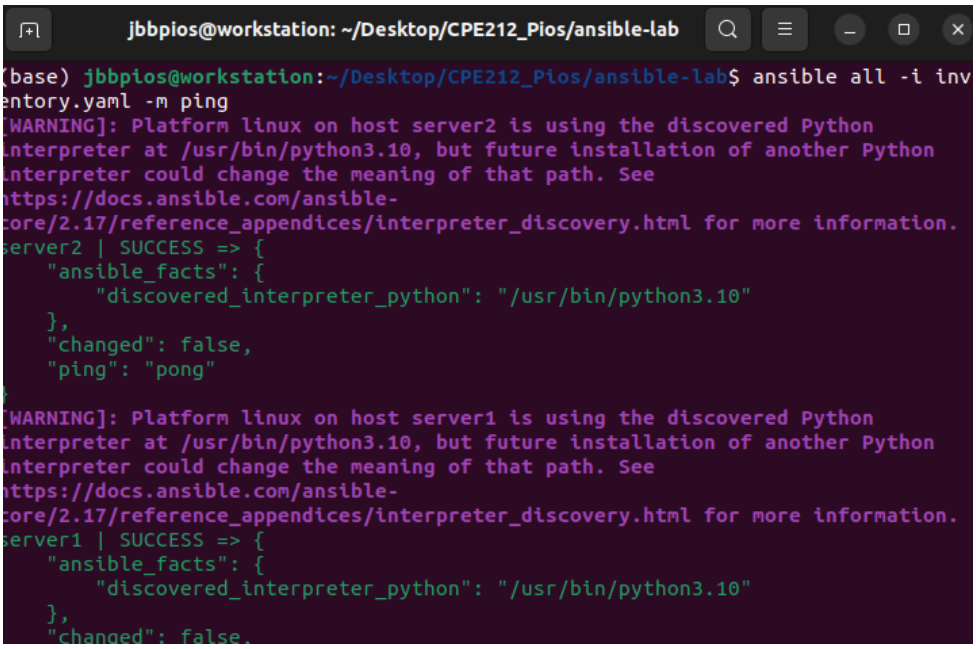
You should be able to see your existing Ansible files and roles

2. Check your existing inventory



Your inventory should contain your previous credentials. You are going to use your previous SSH credentials and will not be hardcoding your sudo password.

3. Test Ansible Connection



Successful Ansible connectivity test to the three managed nodes

4. Install the required Ansible collections

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible-galaxy col
lection install community.general
Starting galaxy collection install process
Nothing to do. All requested collections are already installed. If you want to r
einstall them, consider using `--force`.
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible-galaxy col
lection install ansible.posix
Starting galaxy collection install process
Nothing to do. All requested collections are already installed. If you want to r
einstall them, consider using `--force`.
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$
```

5. Create the Nagios Roles

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ mkdir -p roles/nag
ios_ubuntu/tasks
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ mkdir -p roles/nag
ios_centos/tasks
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$
```

6. Create the Ubuntu Nagios Roles

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ nano roles/nagios_
ubuntu/tasks/main.yml
```

and put this inside:

```
---
- name: Update Ubuntu package cache
  ansible.builtin.apt:
    update_cache: true
    cache_valid_time: 3600

- name: Install Nagios dependencies on Ubuntu
  ansible.builtin.apt:
    name:
      - gcc
      - make
      - autoconf
      - libc6
      - wget
      - unzip
      - apache2
      - php
      - libapache2-mod-php
      - libgd-dev
      - openssl
      - libssl-dev
      - build-essential
```

```
      - snmp
      - libnet-snmp-perl
      - gettext
      - iputils-ping
      - apache2-utils
      - python3-passlib
    state: present

- name: Download Nagios Core
  ansible.builtin.get_url:
    url: https://github.com/NagiosEnterprises/nagioscore/archive/refs/tags/nagios-4.5.13.tar.gz
    dest: /tmp/nagioscore.tar.gz
    mode: '0644'

- name: Extract Nagios Core
  ansible.builtin.unarchive:
    src: /tmp/nagioscore.tar.gz
    dest: /tmp
    remote_src: true
    creates: /tmp/nagioscore-nagios-4.5.13

- name: Create Nagios group
  ansible.builtin.group:
    name: nagios
    state: present

- name: Create Nagios user
  ansible.builtin.user:
    name: nagios
    group: nagios
    shell: /bin/bash
    create_home: true
    state: present

- name: Add Apache user to Nagios group
  ansible.builtin.user:
    name: www-data
    groups: nagios
    append: true

- name: Configure Nagios Core
  ansible.builtin.command:
    cmd: ./configure --with-httpd-conf=/etc/apache2/sites-enabled
    chdir: /tmp/nagioscore-nagios-4.5.13
    creates: /tmp/nagioscore-nagios-4.5.13/Makefile

- name: Compile Nagios Core
```

```
  ansible.builtin.command:
    cmd: make all
    chdir: /tmp/nagioscore-nagios-4.5.13
    creates: /tmp/nagioscore-nagios-4.5.13/base/nagios

- name: Install Nagios Core
  ansible.builtin.command:
    cmd: make install
    chdir: /tmp/nagioscore-nagios-4.5.13

- name: Install Nagios service
  ansible.builtin.command:
    cmd: make install-daemoninit
    chdir: /tmp/nagioscore-nagios-4.5.13

- name: Install Nagios command mode
  ansible.builtin.command:
    cmd: make install-commandmode
    chdir: /tmp/nagioscore-nagios-4.5.13

- name: Install Nagios configuration
  ansible.builtin.command:
    cmd: make install-config
    chdir: /tmp/nagioscore-nagios-4.5.13

- name: Configure Nagios contact email
  ansible.builtin.lineinfile:
    path: /usr/local/nagios/etc/objects/contacts.cfg
    regexp: '^\s*email\s+'
    line: '        email                   nagiosadmin@localhost'

- name: Install Nagios Apache configuration
  ansible.builtin.command:
    cmd: make install-webconf
    chdir: /tmp/nagioscore-nagios-4.5.13

- name: Enable Apache rewrite module
  community.general.apache2_module:
    name: rewrite
    state: present
  notify: Restart Apache

- name: Enable Apache CGI module
  community.general.apache2_module:
    name: cgi
    state: present
  notify: Restart Apache
```

```
- name: Create Nagios web user
  community.general.htpasswd:
    path: /usr/local/nagios/etc/htpasswd.users
    name: nagiosadmin
    password: "{{ nagios_web_password }}"
    owner: nagios
    group: nagios
    mode: '0640'

- name: Allow Apache through UFW
  community.general.ufw:
    rule: allow
    name: Apache

- name: Download Nagios Plugins
  ansible.builtin.get_url:
    url: https://github.com/nagios-plugins/nagios-plugins/archive/refs/tags/release-2.5.tar.gz
    dest: /tmp/nagios-plugins.tar.gz
    mode: '0644'

- name: Extract Nagios Plugins
  ansible.builtin.unarchive:
    src: /tmp/nagios-plugins.tar.gz
    dest: /tmp
    remote_src: true
    creates: /tmp/nagios-plugins-release-2.5

- name: Generate Nagios Plugins configure script
  ansible.builtin.command:
    cmd: tools/setup
    chdir: /tmp/nagios-plugins-release-2.5
    creates: /tmp/nagios-plugins-release-2.5/configure

- name: Configure Nagios Plugins
  ansible.builtin.command:
    cmd: ./configure
    chdir: /tmp/nagios-plugins-release-2.5
    creates: /tmp/nagios-plugins-release-2.5/Makefile

- name: Compile Nagios Plugins
  ansible.builtin.command:
    cmd: make
    chdir: /tmp/nagios-plugins-release-2.5

- name: Install Nagios Plugins
  ansible.builtin.command:
```

```
    cmd: make install
    chdir: /tmp/nagios-plugins-release-2.5
  become: true

- name: Validate Nagios configuration
  ansible.builtin.command:
    cmd: /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
  register: nagios_config_check
  changed_when: false

- name: Enable Nagios
  ansible.builtin.systemd_service:
    name: nagios
    enabled: true

- name: Start Nagios
  ansible.builtin.systemd_service:
    name: nagios
    state: started

- name: Restart Apache
  ansible.builtin.systemd_service:
    name: apache2
    state: restarted
```

7. Create the CentOS role

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ nano roles/nagios_
centos/tasks/main.yml
```

and put this inside:

```
---
- name: Install Nagios dependencies on CentOS
  ansible.builtin.dnf:
    name:
      - gcc
      - glibc
      - glibc-common
      - make
      - gettext
      - gettext-devel
      - autoconf
      - wget
      - unzip
      - httpd
      - php
      - gd
      - gd-devel
```

```
      - openssl
      - openssl-devel
      - net-snmp
      - net-snmp-utils
      - perl-Net-SNMP
      - iputils
      - httpd-tools
    state: present

- name: Download Nagios Core
  ansible.builtin.get_url:
    url: https://github.com/NagiosEnterprises/nagioscore/archive/refs/tags/nagios-4.5.13.tar.gz
    dest: /tmp/nagioscore.tar.gz
    mode: '0644'

- name: Extract Nagios Core
  ansible.builtin.unarchive:
    src: /tmp/nagioscore.tar.gz
    dest: /tmp
    remote_src: true
    creates: /tmp/nagioscore-nagios-4.5.13

- name: Create Nagios group
  ansible.builtin.group:
    name: nagios
    state: present

- name: Create Nagios user
  ansible.builtin.user:
    name: nagios
    group: nagios
    shell: /bin/bash
    create_home: true
    state: present

- name: Add Apache user to Nagios group
  ansible.builtin.user:
    name: apache
    groups: nagios
    append: true

- name: Configure Nagios Core
  ansible.builtin.command:
    cmd: ./configure --with-httpd-conf=/etc/httpd/conf.d
    chdir: /tmp/nagioscore-nagios-4.5.13
    creates: /tmp/nagioscore-nagios-4.5.13/Makefile
```

```
- name: Compile Nagios Core
  ansible.builtin.command:
    cmd: make all
    chdir: /tmp/nagioscore-nagios-4.5.13
    creates: /tmp/nagioscore-nagios-4.5.13/base/nagios

- name: Install Nagios Core
  ansible.builtin.command:
    cmd: make install
    chdir: /tmp/nagioscore-nagios-4.5.13

- name: Install Nagios service
  ansible.builtin.command:
    cmd: make install-daemoninit
    chdir: /tmp/nagioscore-nagios-4.5.13

- name: Install Nagios command mode
  ansible.builtin.command:
    cmd: make install-commandmode
    chdir: /tmp/nagioscore-nagios-4.5.13

- name: Install Nagios configuration
  ansible.builtin.command:
    cmd: make install-config
    chdir: /tmp/nagioscore-nagios-4.5.13

- name: Configure Nagios contact email
  ansible.builtin.lineinfile:
    path: /usr/local/nagios/etc/objects/contacts.cfg
    regexp: '^\s*email\s+'
    line: '        email                   nagiosadmin@localhost'

- name: Install Nagios Apache configuration
  ansible.builtin.command:
    cmd: make install-webconf
    chdir: /tmp/nagioscore-nagios-4.5.13

- name: Check if Nagios web password file exists
  ansible.builtin.stat:
    path: /usr/local/nagios/etc/htpasswd.users
  register: nagios_htpasswd

- name: Create Nagios web user
  ansible.builtin.command:
    cmd: htpasswd -i -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
    stdin: "{{ nagios_web_password }}"
  when: not nagios_htpasswd.stat.exists
```

```
  no_log: true

- name: Update Nagios web user password
  ansible.builtin.command:
    cmd: htpasswd -i /usr/local/nagios/etc/htpasswd.users nagiosadmin
    stdin: "{{ nagios_web_password }}"
  when: nagios_htpasswd.stat.exists
  no_log: true

- name: Set Nagios web password file permissions
  ansible.builtin.file:
    path: /usr/local/nagios/etc/htpasswd.users
    owner: nagios
    group: nagios
    mode: '0640'

- name: Allow HTTP through firewalld
  ansible.posix.firewalld:
    service: http
    permanent: true
    immediate: true
    state: enabled

- name: Download Nagios Plugins
  ansible.builtin.get_url:
    url: https://github.com/nagios-plugins/nagios-plugins/archive/refs/tags/release-2.5.tar.gz
    dest: /tmp/nagios-plugins.tar.gz
    mode: '0644'

- name: Extract Nagios Plugins
  ansible.builtin.unarchive:
    src: /tmp/nagios-plugins.tar.gz
    dest: /tmp
    remote_src: true
    creates: /tmp/nagios-plugins-release-2.5

- name: Generate Nagios Plugins configure script
  ansible.builtin.command:
    cmd: tools/setup
    chdir: /tmp/nagios-plugins-release-2.5
    creates: /tmp/nagios-plugins-release-2.5/configure

- name: Install automake and libtool
  ansible.builtin.dnf:
    name:
      - automake
      - libtool
```

```
    state: present

- name: Generate Nagios Plugins configure files
  ansible.builtin.shell:
    cmd: |
      autoreconf -fiv
    chdir: /tmp/nagios-plugins-release-2.5
    creates: /tmp/nagios-plugins-release-2.5/config.guess


- name: Configure Nagios Plugins
  ansible.builtin.command:
    cmd: ./configure
    chdir: /tmp/nagios-plugins-release-2.5
    creates: /tmp/nagios-plugins-release-2.5/Makefile

- name: Compile Nagios Plugins
  ansible.builtin.command:
    cmd: make
    chdir: /tmp/nagios-plugins-release-2.5

- name: Install Nagios Plugins
  ansible.builtin.command:
    cmd: make install
    chdir: /tmp/nagios-plugins-release-2.5

- name: Validate Nagios configuration
  ansible.builtin.command:
    cmd: /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
  register: nagios_config_check
  changed_when: false

- name: Enable Nagios
  ansible.builtin.systemd_service:
    name: nagios
    enabled: true

- name: Start Nagios
  ansible.builtin.systemd_service:
    name: nagios
    state: started

- name: Enable and start Apache
  ansible.builtin.systemd_service:
    name: httpd
    enabled: true
    state: started
```

8. Create the main playbook that you are going to use

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ nano nagios.yml
```

and put this inside:

```
---
- name: Install Nagios on Ubuntu servers
  hosts: ubuntu_servers
  become: true

  vars_prompt:
    - name: nagios_web_password
      prompt: "Enter Nagios web interface password"
      private: true

  roles:
    - nagios_ubuntu


- name: Install Nagios on CentOS server
  hosts: centos_servers
  become: true

  vars_prompt:
    - name: nagios_web_password
      prompt: "Enter Nagios web interface password"
      private: true

  roles:
    - nagios_centos
```

8. Check the playbook syntax

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible-playbook -
i inventory.yaml nagios.yml --syntax-check

playbook: nagios.yml
```

You should get this with no syntax error

9. Run the Ubuntu group first

Note: the Nagios web interface password is not a pre-existing password. You create it yourself during the ansible installation.

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible-playbook -
i inventory.yaml nagios.yml --limit ubuntu_servers --ask-become-pass
BECOME password:
Enter Nagios web interface password:

PLAY [Install Nagios on Ubuntu servers] ************************************

TASK [Gathering Facts] *****************************************************
[WARNING]: Platform linux on host server1 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]
[WARNING]: Platform linux on host server2 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]

TASK [nagios_ubuntu : Update Ubuntu package cache] *****************************
changed: [server1]
changed: [server2]
```

Your recap should bring you this:

```
PLAY RECAP *********************************************************************
server1                    : ok=28   changed=12   unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
server2                    : ok=28   changed=12   unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
```

10. Verify the Ubuntu servers

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible ubuntu_ser
vers -i inventory.yaml -m shell -a "systemctl is-active nagios" --become --ask-b
ecome-pass
BECOME password:
[WARNING]: Platform linux on host server1 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
server1 | CHANGED | rc=0 >>
active
[WARNING]: Platform linux on host server2 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
server2 | CHANGED | rc=0 >>
active
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$
```

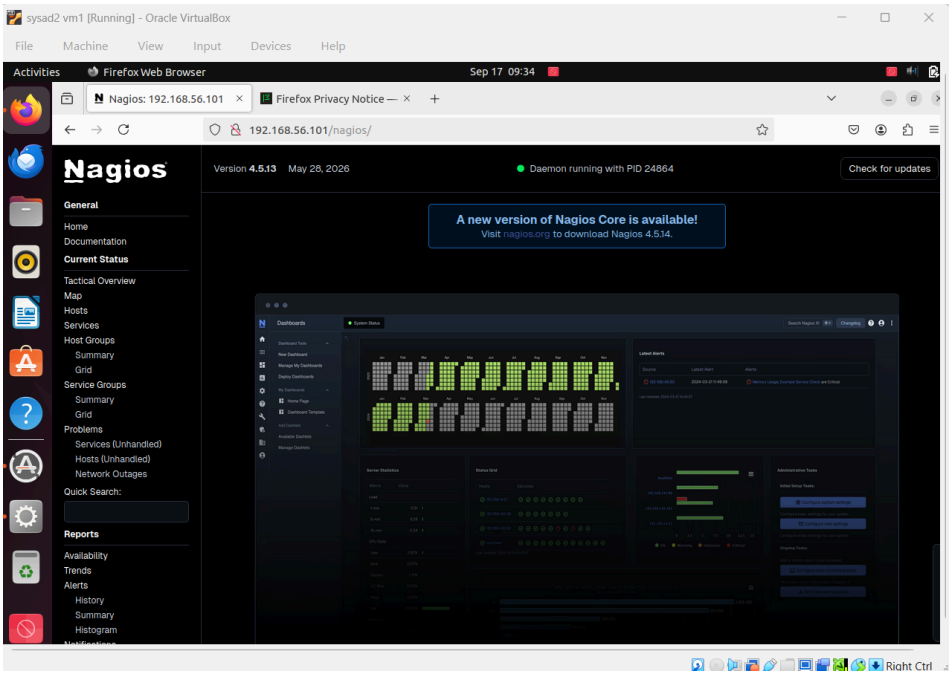
You can also check the configuration:

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible ubuntu_ser
vers -i inventory.yaml -m shell -a "/usr/local/nagios/bin/nagios -v /usr/local/n
agios/etc/nagios.cfg" --become --ask-become-pass
```

```
Total Warnings: 0
Total Errors:   0
```

11. Access the Ubuntu Nagios interfaces

Input this to your browser on your vm1: http://192.168.56.101/nagios
For the login credentials, use:
Username: *nagiosadmin*
Password: *the password you entered during the playbook run*



12. Run the CentOS group

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible-playbook -
i inventory.yaml nagios.yml --limit centos_servers --ask-become-pass
BECOME password:
Enter Nagios web interface password:

PLAY [Install Nagios on Ubuntu servers] ************************************
skipping: no hosts matched
Enter Nagios web interface password:

PLAY [Install Nagios on CentOS server] *************************************

TASK [Gathering Facts] *****************************************************
[WARNING]: Platform linux on host server3 is using the discovered Python
interpreter at /usr/bin/python3.12, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server3]

TASK [nagios_centos : Install Nagios dependencies on CentOS] ***************
```

```
PLAY RECAP *********************************************************************
server3                    : ok=28   changed=10   unreachable=0    failed=0    s
kipped=31   rescued=0    ignored=0
```

13. Check Nagios Service

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible centos_servers -i inventory
.yaml -m shell -a "systemctl status nagios --no-pager" --become --ask-become-pass
BECOME password:
[WARNING]: Platform linux on host server3 is using the discovered Python
interpreter at /usr/bin/python3.12, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
server3 | CHANGED | rc=0 >>
● nagios.service - Nagios Core 4.5.13
     Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: disabled)
     Active: active (running) since Sat 2026-09-19 21:09:10 PST; 23min ago
 Invocation: 621cd85d896f425f9ecc0727f18ad916
       Docs: https://www.nagios.org/documentation
   Main PID: 1158 (nagios)
      Tasks: 6 (limit: 10210)
     Memory: 25.4M (peak: 29.9M, swap: 4K, swap peak: 4K)
        CPU: 1.128s
     CGroup: /system.slice/nagios.service
             ├─1158 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
             ├─1161 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             ├─1162 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             ├─1164 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             ├─1165 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             └─1178 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg

Sep 19 21:09:10 localhost.localdomain nagios[1158]: qh: Socket '/usr/local/nagios/var/rw/nagios.q
h' successfully initialized
Sep 19 21:09:10 localhost.localdomain nagios[1158]: qh: core query handler registered
```

14. Check Apache/httpd

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible centos_servers -i inventory
.yaml -m shell -a "systemctl status nagios --no-pager" --become --ask-become-pass
BECOME password:
[WARNING]: Platform linux on host server3 is using the discovered Python
interpreter at /usr/bin/python3.12, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
server3 | CHANGED | rc=0 >>
● nagios.service - Nagios Core 4.5.13
     Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: disabled)
     Active: active (running) since Sat 2026-09-19 21:09:10 PST; 23min ago
 Invocation: 621cd85d896f425f9ecc0727f18ad916
       Docs: https://www.nagios.org/documentation
   Main PID: 1158 (nagios)
      Tasks: 6 (limit: 10210)
     Memory: 25.4M (peak: 29.9M, swap: 4K, swap peak: 4K)
        CPU: 1.128s
     CGroup: /system.slice/nagios.service
             ├─1158 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
             ├─1161 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             ├─1162 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             ├─1164 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             ├─1165 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             └─1178 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg

Sep 19 21:09:10 localhost.localdomain nagios[1158]: qh: Socket '/usr/local/nagios/var/rw/nagios.q
h' successfully initialized
Sep 19 21:09:10 localhost.localdomain nagios[1158]: qh: core query handler registered
```

15. Verify Nagios Configuration

```
Checking misc settings...

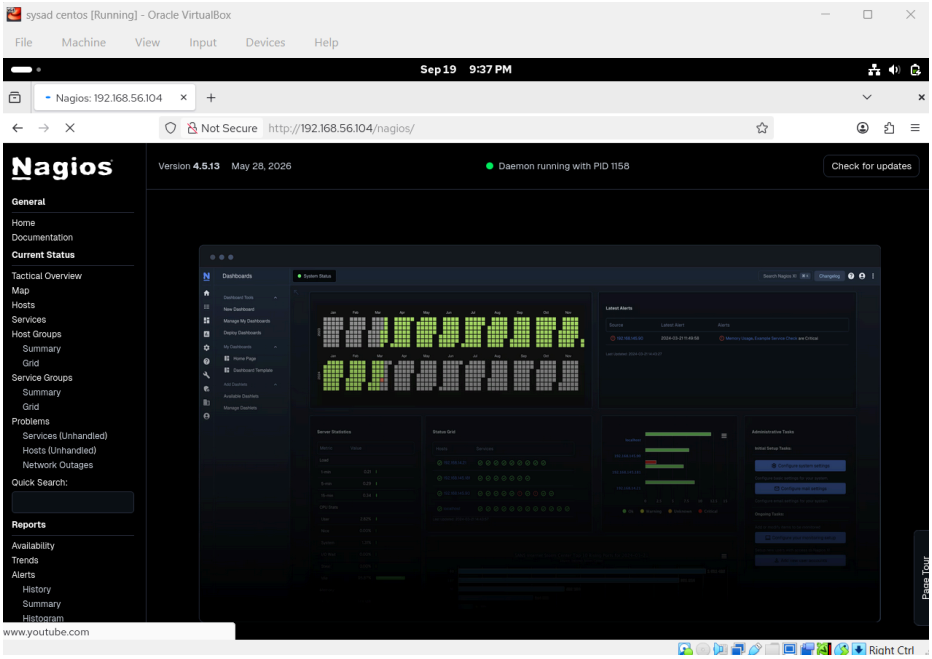
Total Warnings: 0
Total Errors:   0

Things look okay - No serious problems were detected during the pre-flight check
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$
```

16. Check the firewall

```
(base) jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible server3 -i inventory.yaml -
m shell -a "firewall-cmd --list-services" --become --ask-become-pass
BECOME password:
[WARNING]: Platform linux on host server3 is using the discovered Python interpreter at
/usr/bin/python3.12, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
server3 | CHANGED | rc=0 >>
cockpit dhcpv6-client http ssh
```

17. Open Nagios



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. What are the benefits of having an availability monitoring tool?

An availability monitoring tool helps administrators check if servers, applications, and network services are running properly. It can detect problems early, monitor system status, and send alerts when a service becomes unavailable. This helps reduce downtime, makes troubleshooting easier, and ensures that important services remain accessible to users.

Conclusions/Learnings

I learned that availability monitoring is important for maintaining reliable servers and services. Through this activity, I also learned how Nagios Core can be installed and managed using Ansible, which makes the setup more automated and consistent across different Linux servers.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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